Prodigy Anesthesia Quick Review

Autonomic Nervous System

Parasympathetic nerves are supplied via the vagus nerves to what areas of the body?	Esophagus, heart, lungs, entire small intestine, stomach, liver, gallbladder, proximal half of the colon, pancreas, upper portions of the ureters, and the kidneys. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
What beta receptor is responsible for increasing cardiac contractility and increased heart rate?	Beta 1 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
What are the two main subdivisions of the autonomic nervous system?	1. Sympathetic nervous system 2. Parasympathetic nervous system *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
Between what spinal cord segments do sympathetic nerve fibers originate?	T-1 and L-2 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
How do sympathetic nerves differ from skeletal motor nerves?	Skeletal motor nerves are comprised of a single neuron, whereas sympathetic pathways are composed of two neurons which are termed a preganglionic neuron and postganglionic neuron. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
Where are the cell bodies of sympathetic preganglionic neurons found?	In the intermediolateral horn of the spinal cord. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
At what point do preganglionic sympathetic fibers exit spinal nerves?	Immediately following the spinal nerve's exit from the spinal canal. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
Where do sympathetic postganglionic neurons originate?	They originate in one of the peripheral sympathetic ganglia or one of the sympathetic chain ganglia. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
Approximately 75% of all parasympathetic nerve fibers can be found in what cranial nerve?	Cranial nerve 10 (The vagus nerve) *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*

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Parasympathetic nerve fibers located within cranial nerve III innervate what two areas?	The ciliary muscle of the eye and the pupillary sphincter. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
Parasympathetic fibers from cranial nerve VII serve what areas?	The nasal, submandibular, and lacrimal glands. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
Parasympathetic nerve fibers from cranial nerve IX innervate what gland?	The parotid gland *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
Sacral parasympathetic fibers are distributed to what areas of the body?	Descending colon, urinary bladder, rectum, and lower portions of the ureters. They also supply innervation to the external genitalia. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
1. Does the parasympathetic nervous system contain both preganglionic and postganglionic neurons? 2. How do they differ from the sympathetic pathways?	1. Yes. 2. Most preganglionic parasympathetic fibers travel uninterrupted the entire way to the effector organ. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
Where are parasympathetic postganglionic neurons located?	They are located within the wall of the effector organ. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
What alpha receptor results in inhibition of neurotransmitter release when stimulated?	Alpha 2 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
What alpha receptor is responsible for vasoconstriction?	Alpha 1 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
Where do parasympathetic nerve fibers exit the central nervous system?	Through cranial nerves 3, 7, 9 and 10 and the 2nd and 3rd sacral spinal nerves. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 730.*
What beta receptor is responsible for glycogenolysis, lipolysis, and bladder wall relaxation?	Beta 2 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*

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Autonomic Nervous System

Are the sympathetic nerve fibers that terminate in the adrenal medulla preganglionic or postganglionic? What neurotransmitter do they secrete?	They are preganglionic, and therefore secrete acetylcholine. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 369.*
What beta receptor is responsible for uterine and intestinal relaxation?	Beta 2 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
What beta receptors are located in the lungs that result in bronchodilation when stimulated?	Beta 2 receptors. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
What beta receptor is responsible for thermogenesis?	Beta 3 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
Name the two main receptors that acetylcholine activates.	1. Nicotinic receptors 2. Muscarinic receptors *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
1. The majority of sympathetic postganglionic neurons are adrenergic, and therefore secrete what substance? 2. Sympathetic postganglionic nerve fibers serving what three areas are cholinergic?	Norepinephrine 2. Nerve fibers to the piloerector muscles of the hairs, sweat glands, and a small number of blood vessels are cholinergic. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 731.*
Almost all of the parasympathetic postganglionic neurons are cholinergic, and therefore secrete what neurotransmitter?	Acetylcholine *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 731.*
All preganglionic neurons (sympathetic and parasympathetic) are cholinergic, and therefore secrete what neurotransmitter?	Acetylcholine *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 368.*
Are nerve fibers that secrete norepinephrine considered cholinergic or adrenergic?	Adrenergic *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 731.*

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Are nerve fibers that secrete acetylcholine considered cholinergic or adrenergic?	Cholinergic *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 731.*
What are the two major types of adrenergic receptors?	1. Alpha receptors 2. Beta receptors. These receptors are further divided into alpha 1 and alpha 2 receptors, as well as beta 1, beta 2, and beta 3 receptors. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
How does stimulation of the sympathetic nervous system affect sweat glands?	It causes them to release copious amounts of sweat. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does sympathetic stimulation affect the detrusor and trigone muscles of the bladder.	There is relaxation of the detrusor muscle and contraction of the trigone muscle. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What are the renal effects of sympathetic nervous system stimulation?	There is decreased urine output and an increase in the secretion of renin. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What is the effect of sympathetic stimulation on the gallbladder and bile ducts?	It causes them to relax. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What effect does sympathetic stimulation have on the liver?	It causes the liver to release glucose. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect the gut?	It causes a reduction in peristalsis and decreased tone within the lumen. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect the lungs?	It causes dilation of the bronchi and mild constriction of blood vessels. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What happens when beta 2 receptors within the coronary arteries are stimulated?	The coronary arteries dilate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*

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What is the affect of sympathetic nervous system activation on heart muscle?	There is an increase in the force of contraction of heart muscle. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the sympathetic nervous system affect the pupil?	It dilates the pupil. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the sympathetic nervous system affect nasal, parotid, submandibular, gastric, pancreatic, and nasal glands?	It causes vasoconstriction and minimal secretion. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the sympathetic nervous system affect apocrine glands?	It causes them to release odiferous, thick secretions. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
Between what spinal cord segments do preganglionic sympathetic nerve fibers originate?	The preganglionic sympathetic nervous system fibers originate between the T-1 and L-2 nerve roots. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
How does activation of the sympathetic nervous system affect coagulation?	It causes an increase in coagulation. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect the penis?	It causes ejaculation. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect heart rate?	It causes an increase in heart rate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does sympathetic nervous system activation affect basal metabolism?	It increases it up to 100% *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does sympathetic nervous system activation affect skeletal muscle?	There is an increase in glycogenolysis and an increase in strength. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*

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What effect does activation of the sympathetic nervous system have on fat cells?	It causes lipolysis. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect adrenal medulla secretion?	It causes an increase. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the sympathetic nervous system affect systemic arterioles of the abdominal viscera and skin?	It causes them to constrict. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the sympathetic nervous system affect the piloerector muscles?	It causes them to contract. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect heart rate?	It produces a reduction in heart rate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect the penis?	It causes erection of the penis. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the parasympathetic nervous system affect lacrimal, parotid, nasal, submandibular, pancreatic, and gastric glands?	It causes increased secretion in each of these glands. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect adrenal medulla secretion?	There is no effect. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect sweat glands?	It produces sweat on the palms of hands. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
Does activation of the parasympathetic nervous system have any effect on skeletal muscle?	No *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*

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Does stimulation of the parasympathetic nervous system have any effect on apocrine glands?	No. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the parasympathetic nervous system affect the pupil?	It constricts the pupil. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
Is there any parasympathetic activity in fat cells?	No *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
Is there any parasympathetic effects on coagulation?	No. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does parasympathetic stimulation affect the detrusor and trigone muscles of the bladder?	There is contraction of the detrusor muscle and relaxation of the trigone muscle. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does parasympathetic stimulation affect the coronary arteries?	It causes dilation. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect bronchi within the lungs?	It causes the bronchi to constrict. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does parasympathetic nervous system activation affect basal metabolism?	There is no effect. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect the gut?	It increases the muscle tone of the gastrointestinal tract and an increase in peristalsis. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does parasympathetic activation affect the liver?	It causes a slight increase in glycogen synthesis. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*

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Autonomic Nervous System

What is the effect of parasympathetic stimulation on the gallbladder and bile ducts?	It causes them to contract. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does stimulation of the parasympathetic nervous system affect the piloerector muscles?	There is no effect. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
How does activation of the parasympathetic nervous system affect the kidney?	There is no effect. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
Does activation of the parasympathetic nervous system affect systemic arterioles?	There is no effect. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What is the affect of parasympathetic nervous system activation on heart muscle?	There is a decreased force of contraction of heart muscle. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 734.*
What two hormones are released into the bloodstream in mass quantities when stimulation of sympathetic nerves to the adrenal medulla occurs?	Epinephrine and norepinephrine *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 736.*
What are the two divisions of the Autonomic Nervous System?	The two divisions of the Autonomic Nervous System are the Sympathetic (Adrenergic) and the Parasympathetic (Cholinergic) Nervous Systems *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 363.*
Where are muscarinic receptors located?	They are located on all effector cells stimulated by postganglionic cholinergic neurons of either the PNS or SNS *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*
What are the two main types of receptors that acetylcholine activates?	1. Muscarinic 2. Nicotinic *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 733.*

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Autonomic Nervous System

Cardiac pump function is affected by the SNS and PNS in what three ways?	1. Changing the strength of the contraction (inotropism) 2. Changing the rate (chronotropism) 3. The modulation of coronary blood flow *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 362.*
Are postganglionic neurons of the SNS considered to be cholinergic or adrenergic?	Adrenergic. The exception to this are postganglionic sympathetic nerve fibers to the piloerector muscles of the hairs, sweat glands, and a few blood vessels. These nerve fibers are considered to be cholinergic *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 721.*
Are postganglionic neurons of the PNS considered to be cholinergic or adrenergic?	Cholinergic *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 721.*
Are preganglionic neurons in the sympathetic and parasympathetic nervous system cholinergic or adrenergic?	All preganglionic neurons in both the SNS and the PNS are cholinergic, and therefore secrete acetylcholine. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 368.*
What is the primary neurotransmitter of the PNS?	Acetylcholine *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 368.*
Where do sympathetic nerve fibers originate?	In the spinal cord and spinal nerves between the cord levels of T1 and L2. From there, they pass into the sympathetic chain and then on to the organs and tissues they innervate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
The ANS is responsible for regulating what functions?	This system helps with the control of arterial pressure, gastrointestinal secretion, urinary bladder emptying, gastrointestinal motility, sweating, body temperature, and involuntary regulation of smooth and cardiac muscle. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 729.*
All sympathetic neurotransmitters are synthesized from what substance?	All sympathetic neurotransmitters are synthesized from tyrosine. The synthesis takes place in the postganglionic sympathetic nerve ending. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 69.*

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